Tejesvani Muppara Vijayaram

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OBJECTIVE

Data Engineer with 2 years of experience in designing ETL pipelines, building scalable data workflows, and developing interactive dashboards. Proficient in Python, SQL, Tableau, and cloud platforms (AWS, GCP), with expertise in data processing and predictive analytics. Passionate about driving business efficiency through data-driven solutions.

EDUCATION

Northeastern University, Boston — **Sept 2023 - May 2025**
MS Data Analytics Engineering — *CGPA 3.94*

Anna University, Tamil Nadu, India — **Sept 2017 - May 2021**
BE Computer Science and Engineering — *CGPA 3.64*

TECHNICAL SKILLS

Programming Languages: Python (NumPy, Pandas, Seaborn, Matplotlib, Scikit-learn), SQL
ETL & Analytical Tools: Tableau, Power BI, Looker, Apache Airflow, Kafka, MS Excel (VLOOKUP, Pivot Tables, VBA), Spark
Data Warehouse & Cloud: MySQL, AWS (S3, EC2, Glue, Athena, Redshift), GCP, BigQuery, Snowflake
Developer & Automation Tools: Git, GitHub Actions, Docker, Jira

WORK EXPERIENCE

Data Engineer — **Aug 2021 – May 2023**
LTIMindtree — *Chennai, IN*

- Optimized **ETL pipelines** to migrate 2M+ policyholder records from **Oracle DB** to **AWS**, utilizing **AWS DMS** and **Redshift**, improving data accessibility and analytics capabilities.
- Implemented data workflows integrating **Amazon RDS** with **AWS Redshift** via **AWS Glue**, reducing processing time 6 hours, while enabling **advanced analytics** on claims processing and policyholder behavior, and **ad-hoc reporting** on demand.
- Automated data validation checks using **Airflow DAGs** and **SQL scripts**, ensuring data integrity and data quality across 500K records and reducing manual verification by 40%.
- Increased customer retention by 5% by leveraging **Tableau dashboards** to track key insurance **KPIs**, including **claims processing efficiency, policyholder retention, and underwriting accuracy**, driving data-driven decision-making.
- Led **requirement analysis** to devise technical solutions, reducing post-release defects by 15% via optimized implementation.
- Collaborated with cloud engineers to optimize workflows, enhancing operational efficiency and improving analytics scalability.

PROJECTS

IoT Data Streaming and Analytics (Project Code)

- Built a data streaming pipeline using **Kafka** and **Spark** to process data from 4 IoT sensors(Vehicle, GPS, Weather, Emergency).
- Refined ETL process with **AWS Glue** for data transformation and cataloging, increasing data processing efficiency by 35%.
- Integrated **Amazon S3** and **Redshift** for data storage and optimization, leveraging **Athena** for efficient querying and analytics.

Bank Market Prediction Analysis (Project Code)

- Developed a scalable **MLOps pipeline** using **Random Forest** to predict **term deposits** from **50K+ financial records**.
- Automated data preprocessing, validation, and model building using **Airflow** and **Docker**, reducing manual effort by 70%.
- Integrated **GitHub Actions** for **CI/CD** and **MLflow** for model tracking, and deployed the model using Flask on GCP with real-time monitoring of data drift, while achieving an accuracy of 84% of term deposit predictions.

Bike Sharing Demand Analysis (Project Code)

- Engineered **predictive model** for bike-sharing demand, improving forecast accuracy by 8% and optimizing fleet management.
- Calibrated the **Random Forest** model with **GridSearchCV**, reducing MAE by 10% and increasing R-squared by 2.3%.
- Leveraged statistical methods like **t-tests** and **chi-squared tests** to validate the impact of weather, seasons, and holidays on bike demand, uncovering key usage patterns driving bike-sharing usage and operations.

Sales and Customer Dashboard (Dashboard Link)

- Transformed a 13K-record retail dataset into a **STAR schema** using **Python**, optimizing data structure for efficient analysis.
- Crafted **Tableau dashboards** to analyze KPI, including **sales** and **profit metrics**, facilitating data-driven decision-making.
- Designed the dashboards with **advanced filters**, uncovering customer behavior, sales trends, and reducing stock-outs by 21%.

CERTIFICATIONS

AWS Certified Cloud Practitioner - AWS
Programming, Data Structures and Algorithms using Python - NPTEL
Google Data Analytics Professional Certificate - Coursera